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Spam SMS (or) Email Detection and Classification using Machine Learning

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***Abstract*—Spam is an unwanted message or SMS sent on mobile phones whose content may be malicious. Scammers send fake text messages to trick people into responding to their SMS and they may hack personal information, password, account number, etc. To avoid being tricked by scammers, proposed a model based on Machine learning Algorithms. The proposed model is implemented using the Naïve Bayes algorithm and term frequency-inverse document frequency vectorizer. Obtained the dataset from Kaggle and trained the model using it. This model consists of a local host website which is obtained through PyCharm IDE. Obtained results show that the model accuracy of 95% and a precision of 100%.**

Keywords— Spam SMS, Spam Email, Machine Learning, Naïve Bayes, Cyber Crime, Cyber Scam

I. INTRODUCTION

The whole world is moving towards digitalization. People converse, send money and do many activities which make life easier. Even though it has many pros it also has many cons too. Nowadays people are targeted by online scammers and get tricked easily. People may receive suspicious links, unrecognized contact numbers, offers, etc. through emails, SMS, and social media. The messages can be received randomly or targeted on particulars. Sometimes the messages might seem to be non-spam which can trick people and can get successful in scamming. Online scams come under cybercrime and the thief can be sentenced to punishments but due to a lack of awareness in the general public, these crimes can go unnoticed which may promote these scam activities more. Cybercrime offices, telecom companies, and banks warn people about spammers and hackers who trick people by sending messages, links, and emails. But normally people are not aware of whether the messages and emails they get are verified or fake due to this reason cyber scams happen[19]. A private firm named Local Circles conducted a survey in which the statistics showed that in the last 3 years 42% of Indians faced financial fraud and 74% of people failed to retrieve the money. To overcome these cyber scams, proposed a model based on machine learning which helps individuals to check if the messages and emails they are receiving are spam or not. Whenever the user feels the message is unsafe, they can copy and paste it into the open-source site created.

II. LITERATURE REVIEW

Lutfun Nahar Lota *et al.* [1] explained how SMS spam detection can be more challenging compared to other types of detection. Different algorithms that can be used and the scope of improvisation. Through this paper, able to study many more literature reviews on the same topic. Paras Sethi *et al*. [2] explained how big of a problem SMS spam is globally. different algorithms available for analyzing the model and which ones can be the best among them and also studied different filtering methods. Shafi'i muhammad abdulhamid1 *et al*. [3] summarized a review of the methods that are generally used, probable challenges that can occur, and the scope for future research on spam filtering and spam detection in mobile SMS. This paper provided the details on available filtering techniques that can be used in the model to check different parameters like accuracy. M.Rubin julis *et al.* [4] explained how to cleanse the dataset, different phases of data cleaning, how to analyze the dataset and represent them, and several filtering techniques. Amani Alzahrani *et al*. [5] explained how to represent the cleaned data by editing column names and representing the analyzed data through different graphs for different parameters, representing the precision and accuracy of each model used for analysis. Naina Nisar *et al*. [6] elaborated on different classifiers like SVM, NB, k-NN, DT, RF, and Adaboost along with the analysis. Studied Voting Ensemble which is applied in the training phase of model to check the accuracy and precision. Sakshi Agarwal *et al*. [7] proposed the addition of Indian spam messages to the available dataset acquired by searching and collecting data globally and analyzing them. This study has reviewed various methods and phases of SMS spam detection along with different classifiers that can be used in model training. Michael Crawford *et al*. [8] described a practice known as opinion review spam. In this, spammers give fake reviews that they manipulate people through reviews. They give fake, untruthful, and deceptive reviews to gain the trust of people. proposed method using natural language processing (NLP). opinioned that one's need to understand that all the reviews may not be genuine and reliable, there is a requirement to develop a technique to detect spam and review that. Anju Radhakrishnan *et al*. [9] summarized different machine learning techniques for filtering spam emails using different SMS datasets or email datasets. Nikhil Govil *et al*. [10] elaborated on the steps involved in building the model in detail. This paper describes the generation and testing of the machine learning model, spam filtering algorithm, spam identification methods, and Naïve Bayes algorithms. Shirani-Mehr *et al*. [11] explained that since SMS spam detection is a fairly new area and the techniques are in the development stage, there may be a lack of reliable reviews. Elaborated on how they have considered different literature reviews and gone through different algorithms to come up with a solution of great accuracy. Fette, I *et al*. [12] described how necessary standard the world wide web has become from the point of view of securing information and knowledge. It is used to identify hackers and malicious users. Explained how emails are used as phishing tools. Shaufiah *et al*. [13] described how obtaining accuracy

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is the most challenging part of the process of SMS spam filtering, combining the Naïve Bayes classifier and FP-growth algorithm in order to enhance the performance in classifying spam and ham SMS. Andronicus A *et al*. [14] elaborated on the major usage of random forest algorithms in the application of detection of fraudulent attempts through emails and have also developed prediction accuracy by training with a dataset containing 2000 phishing and ham emails. They got the resulting classification accuracy of 99.7%. G. Tripathi *et al*. [15] described the application of machine learning algorithms and natural language processing in the proposed model for computational analysis. Here the behavior of classifiers is observed that is Naïve Bayes and SVM.

III. DESIGN AND IMPLEMENTATION

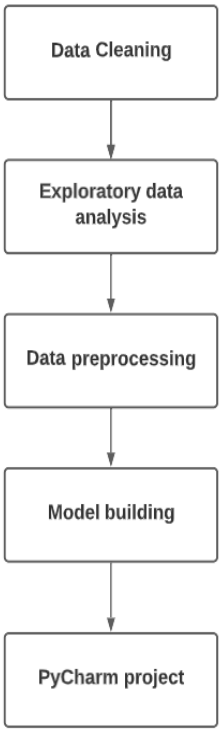


Fig 1. Block Diagram of spam SMS detection

Figure 1 shows the block diagram of spam SMS detection. The major steps in the process are data cleaning, exploratory data analysis, data preprocessing, model building, and the PyCharm IDE. Figure 2 shows the detailed flowchart of spam SMS detection. Initially, the dataset is separated as spam and ham. Spam is assigned with 1 and ham is assigned with 0. Each SMS from the dataset breaks into a number of characters, words, and sentences and then be analyzed. The data converted to lowercase, tokenized, and stemming will be applied. Using Naïve Bayes algorithms, it is vectorized. Hence, this leads to the completion of model building. Used Jupyter Notebook, Jupyter is an open-source, web-based platform that permits you to work in all languages. It provides the service to work, create, interact, edit, share, analyze and arrange documents. Installed and imported the required python libraries into Jupyter Notebook.

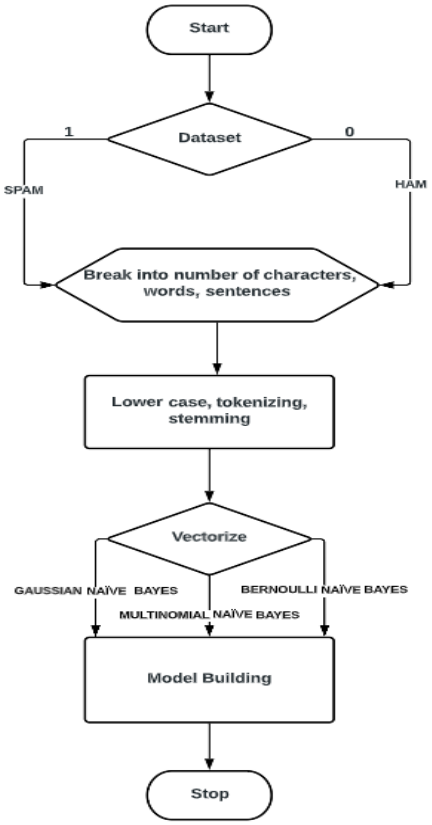


Fig 2. Flowchart of spam SMS detection

Data Cleaning: This stage consists of dropping the columns which are unused from the dataset and then assigning values 1 for spam messages and 0 for ham messages. Check if there are any missing values are present or not in the data set and Check for duplicate values in the data set if any are present then we have to remove the duplicates by dropping them.

EDA (Exploratory Data Analysis): Here we check how many messages are spam and how many messages are ham in percentage and represent that in a pie chart. For deeper analysis, we make three columns for the number of characters, words, and sentences in the messages. Analyze characters, words, and sentences of spam and ham messages separately and also plot and analyze the correlation coefficient.

Data Preprocessing: In this process, the conversion of all the letters into lowercase letters takes place. Then tokenizing all the sentences. Removal of all stop words, special characters, and punctuation. Apply stemming on all the sentences. Using WordCloud analyzes the most used words in spam and ham.

Model Building: Here, vectorize all the words using tfidf. Then train and test using Gaussian Naïve Bayes, Multinomial Naïve Bayes, and Bernoulli Naïve Bayes. From this, the accuracy and precision of the above-mentioned methods.

By importing pickle, get two files vectorizer.pkl and model.pkl. Further created an IDE in PyCharm. Uploaded pickle files in PyCharm. To create a website, used streamlit, an open-source python library. After writing code in PyCharm get a local URL as output. On the website, get a block to enter the message and it has a predict button that informs us whether the message is spam or ham.

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IV. DATASET SPECIFICATIONS

Figure 3 shows the dataset specifications, downloaded from Kaggle, an online community. The dataset consists of 5570 spam and ham messages, out of which 12.63% SMS are spam and 87.37% SMS are not spam. In the data cleaning process duplicate and unlabeled messages have been removed. Analyzed the data and visualized it through a pie chart as spam and ham. Further, analyzed the dataset and removed duplicate data and unlabeled data.

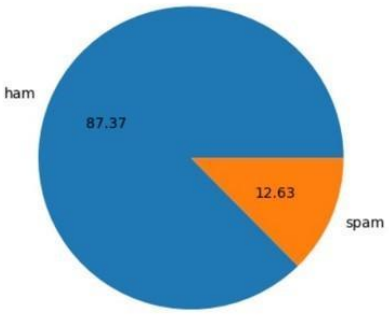


Fig 3. Dataset classifying spam and ham

V. SIMULATION RESULTS AND ANALYSIS

Steps in building an SMS spam detection model are, data from the dataset needs to be cleaned first. Then it has to go through exploratory data analysis where we analyze the data considering various parameters like characters, words, sentences, etc. In the data preprocessing conversion, tokenization, removal, and stemming happen. Then it is ready for the model building where vectorize it. Then import files to PyCharm and build the model to get the desired output. The dataset consists of 87.37% ham SMS and 12.63% spam SMS. The heatmap shows the correlation. Here in Fig 4., 1 being spam and 0 being ham it shows the correlation of the number of characters, number of words, and number of sentences with all the combinations.

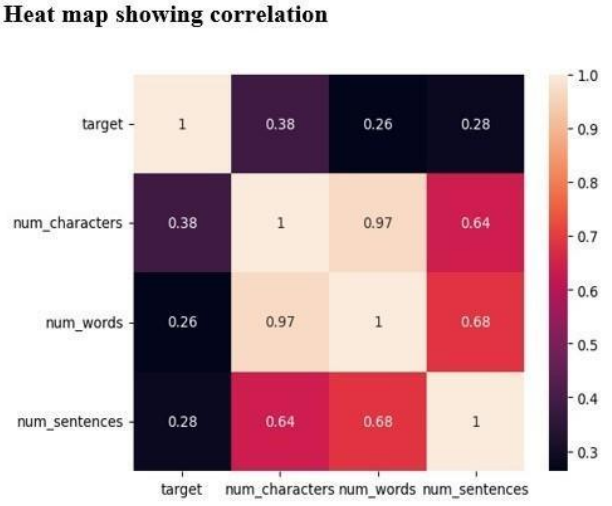


Fig 4. Heat map showing correlation

The heat map shows that as the number of characters increases the tendency to be spam increases. Since the variation of the number of characters with the target is 0.38, we chose it for the model. A correlation heatmap is a matrix showing 2D correlation between two discrete dimensions using colored cells to represent data. There are steps to interpret heatmap correlation since the value of correlation can take values from -1 to 1, if 1 it means that it is a positive correlation if -1 it means it is a negative correlation and if zero it means there is no correlation. Analyzed the dataset separately for spam and SMS and visualized the most repeated words with the number of frequencies in which they are being repeated.

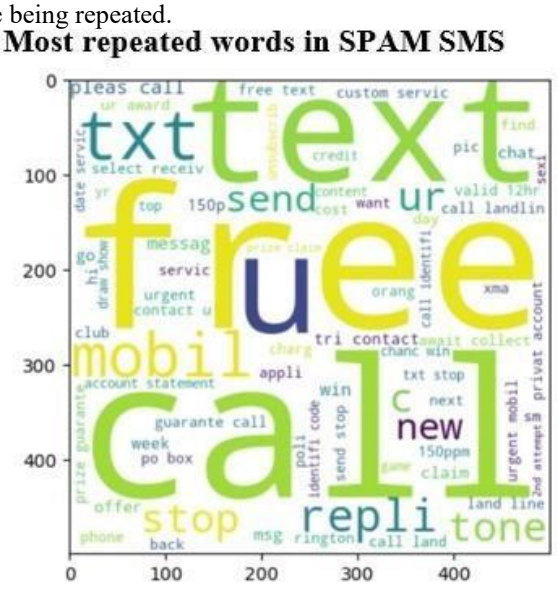


Fig 5. Most repeated words in spam SMS

Most repeated words in the spam SMS are shown in figure 5 bigger letters. Text, free, call, u, etc. are a few of the most repeated words**.** This can be found by using Word Clouding. Word cloud is a visual representation of a text, in which the words appear bigger the more often they are mentioned. Considering the data set these are the most often used words in spam messages.



Fig 6. Most repeated words in ham SMS

Most repeated words in the ham SMS are shown in figure 6 bigger letters. Go, got, come, etc. are a few of the most repeated words in the not-spam SMS. This is also found out by using the word cloud where the bigger words represent the frequency of the word used. This helps to identify if the given message is spam or ham.

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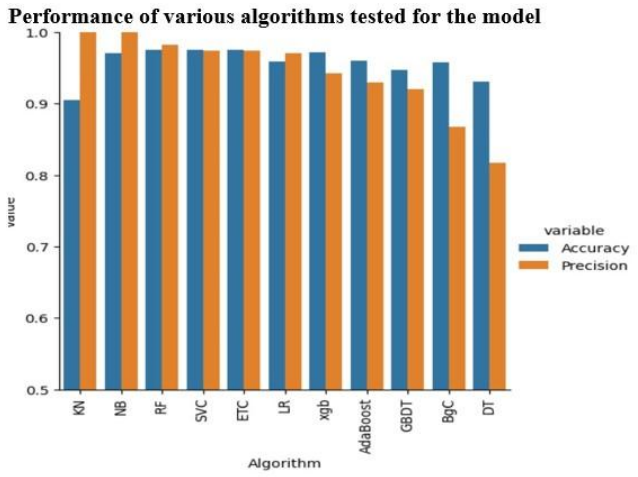


Fig 7. Model comparison based on performance

Figure 7. Bar graph shows the accuracy and precision values of all the algorithms used in the model-building process. KN, Naïve Bayes, Random Forest, SVC, ETC, Linear regression, etc. are the algorithms used. Through this analysis, decided usage of which algorithm will be suitable and bring good results is decided. As needed a good amount of both accuracy and precision, Accuracy measures how close results are to the true or known value. Precision, on the other hand, measures how close results are to one another. They're both useful ways to track and report on results. first compared the performances of each algorithm with one another. Finally chosen Naïve Bayes as it outperformed the other algorithms.

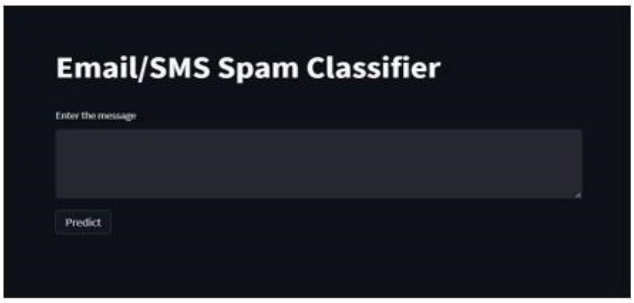


Fig 8. Email/SMS spam classifier window

Figure 8 shows the Email/SMS spam classifier window, imported files from Jupyter are uploaded in PyCharm. On running a block of code in the PyCharm panel, the output leads to a local host where can see a block to paste the SMS to check whether the SMS is spam or not, and along with that, a predict button is provided.



Fig 9. The result showing spam SMS

Figure 9 shows the result of spam SMS. In the provided block named enter the message, paste the SMS message which is received to check whether the message is spam or ham.



Fig 10. The result showing ham SMS

Figure 10 shows the result of ham SMS. Upon clicking predict button it shows provided SMS is spam or not spam. This is a process to be followed to check whether the message is spam or not spam. The process is fairly comfortable with easy going and gives good results. This work can be deployed into an app.

VI. EMAIL SPAM DETECTION

The proposed model can also be implemented for email spam detection. Since Emails are lengthier than SMS it needs a much longer and bigger dataset. Email spam detection can be easier than SMS spam detection due to the length of the text, mentioned clear intention, possible availability of the sender information, and awareness about spam Emails.

VII. REAL TIME IMPLEMENTATION CHALLENGES

Having an insufficient dataset causes a lack of accuracy and precision. Having a smaller number of spam message samples might be a disadvantage. Depending on the dataset, different machine learning algorithms show different results and therefore require the application of different algorithms and testing to get the desired result. Implementing this model in real-time would be quite challenging and therefore might need the usage of better technology or software. Deploying this model into an app is challenging. Since the messaging apps don't allow access of content to the spam detection app or website, users have to copy and paste them manually to check their credibility. Smart spammers and hackers might come up with creative ways of messaging which makes it harder to identify the spam for the website or app.

VIII. APPLICATIONS

SMS is one of the most accessible ways of communication mediums all over the world. The SMS feature is easy to use, and it works without the internet. Therefore, along with more users, it also gains more scammers. Since SMS doesn't have much security, it is easy for scammers to send the same SMS to many people. SMS can be used by people who aren't aware of technologies, safety issues, and online frauds that may lead to people getting scammed. Users may get unwanted messages which might be irritating, irrelevant ads, fake job offers, lotteries, contact information, and links that upon clicking lead to scams. Scammers may get money, contact information, address, private information, etc. which may or may not come under cybercrimes. Online scams may expose users to serious dangers. The lack of security systems and the unawareness of the users make scamming easier**.** Therefore, to prevent SMS users from slipping into the dangers of scams

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and fraud the above model is proposed. Whenever the user receives any suspicious, random, unknown SMS they can make use of this feature and get to know the credibility[16][17][18]. The model helps in raising awareness among users about spam SMS and makes it easier to recognize spam.

IX. CONCLUSION

The danger of spam SMS is increasing all over the world at a very high rate and keeps on accelerating since access to the internet and mobile connectivity has increased. India is getting higher exposure to this phenomenon because of the availability of SMS services at lower cost. As a matter of precaution and to avoid fraud occurrences the model proposes a machine learning-based solution. The presented spam SMS filtering method is analyzed based on various algorithms, visualized through graphs and charts, and finally based on performance, accuracy, and precision; it implements TF-IDF with the Naïve Bayes classification. The proposed model is a website consisting of a block to write a message and a prediction button that informs whether the message is spam or not. This makes the model easy to use and adaptable for all age groups of people. As this model gives accuracy and precision of more than 95%, to protect ourselves from most online scams.

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